



Fever and Antipyretic Use in Children

Janice E. Sullivan, MD, Henry C. Farrar, MD, and the SECTION ON CLINICAL PHARMACOLOGY AND THERAPEUTICS, AND COMMITTEE ON DRUGS

Clinical Report – Reaffirmed With Reference & Data Updates

This clinical report was reaffirmed in October 2022 with reference and data updates. New or updated references and datapoints are indicated in bold typeface. No other changes have been made to the text or content. The AAP would like to acknowledge John W. Berkenbosch, MD, FAAP, FCCM, for these updates.

abstract

Fever in a child is one of the most common clinical symptoms managed by pediatricians and other health care providers and a frequent cause of parental concern. Many parents administer antipyretics even when there is minimal or no fever, because they are concerned that the child must maintain a “normal” temperature. Fever, however, is not the primary illness but is a physiologic mechanism that has beneficial effects in fighting infection. There is no evidence that fever itself worsens the course of an illness or that it causes long-term neurologic complications. Thus, the primary goal of treating the febrile child should be to improve the child’s overall comfort rather than focus on the normalization of body temperature. When counseling the parents or caregivers of a febrile child, the general well-being of the child, the importance of monitoring activity, observing for signs of serious illness, encouraging appropriate fluid intake, and the safe storage of antipyretics should be emphasized. Current evidence suggests that there is no substantial difference in the safety and effectiveness of acetaminophen and ibuprofen in the care of a generally healthy child with fever. There is evidence that combining these 2 products is more effective than the use of a single agent alone; however, there are concerns that combined treatment may be more complicated and contribute to the unsafe use of these drugs. Pediatricians should also promote patient safety by advocating for simplified formulations, dosing instructions, and dosing devices.

The guidance in this report does not indicate an exclusive course of treatment or serve as a standard of medical care. Variations, taking into account individual circumstances, may be appropriate.

All clinical reports from the American Academy of Pediatrics automatically expire 5 years after publication unless reaffirmed, revised, or retired at or before that time.

DOI: <https://doi.org/10.1542/peds.2010-3852>

PEDIATRICS (ISSN Numbers: Print, 0031-4005; Online, 1098-4275).

Copyright © 2011 by the American Academy of Pediatrics

This document is copyrighted and is property of the American Academy of Pediatrics and its Board of Directors. All authors have filed conflict of interest statements with the American Academy of Pediatrics. Any conflicts have been resolved through a process approved by the Board of Directors. The American Academy of Pediatrics has neither solicited nor accepted any commercial involvement in the development of the content of this publication.

INTRODUCTION

Fever is one of the most common clinical symptoms managed by pediatricians and other health care providers and accounts, by some

To cite: Sullivan JE, Farrar HC, et al; Section on Clinical Pharmacology and Therapeutics. Fever and Antipyretic Use in Children. *Pediatrics*. 2011;127(3):e20103852

estimates, for one-third of all presenting conditions in children.¹ Fever in a child commonly leads to unscheduled physician visits, telephone calls by parents to their child's physician for advice on fever control, and the wide use of over-the-counter antipyretics.

Parents are frequently concerned with the need to maintain a "normal" temperature in their ill child. Many parents administer antipyretics even though there is either minimal or no fever.² Approximately one-half of parents consider a temperature of less than 38°C (100.4°F) to be a fever, and 25% of caregivers would give antipyretics for temperatures of less than 37.8°C (100°F).^{1,3,4} Furthermore, 85% of parents (*n* = 340) reported awakening their child from sleep to give antipyretics.¹ Unfortunately, as many as one-half of parents administer incorrect doses of antipyretics; approximately 15% of parents give supratherapeutic doses of acetaminophen or ibuprofen.⁵ Caregivers who understand that dosing should be based on weight rather than age or height of fever are much less likely to give an incorrect dose.⁵

Physicians and nurses are the primary source of information on fever management for parents and caregivers, although there are some disparities between the views of parents and physicians regarding antipyretic treatment.⁶ The most common indications for initiating antipyretic therapy by pediatricians are a temperature higher than 38.3°C (101°F) and improving the child's overall comfort.⁷ Although only 13% of pediatricians specifically cite discomfort as the primary indication for antipyretic use,⁸ this intent is generally implied in their recommendations. Most pediatricians (80%) believe that a sleeping ill child should not be

awakened solely to be given antipyretics.⁷

Antipyretic therapy will remain a common practice by parents and is generally encouraged and supported by pediatricians. Thus, pediatricians and health care providers are responsible for the appropriate counseling of parents and other caregivers about fever and the use of antipyretics.^{9,10}

PHYSIOLOGY OF FEVER

It should be emphasized that fever is not an illness but is, in fact, a physiologic mechanism that has beneficial effects in fighting infection.^{11–13} Fever retards the growth and reproduction of bacteria and viruses, enhances neutrophil production and T-lymphocyte proliferation, and aids in the body's acute-phase reaction.^{14–17} The degree of fever does not always correlate with the severity of illness. Most fevers are of short duration, are benign, and may actually protect the host.¹⁸ Data show beneficial effects on certain components of the immune system in fever, and limited data have revealed that fever actually helps the body recover more quickly from viral infections, although the fever may result in discomfort in children.^{14,19–21} Evidence is inconclusive as to whether treating with antipyretics, particularly ibuprofen alone or in combination with acetaminophen, increases the risks of complications with certain types of infections.^{22,23} Potential benefits of fever reduction include relief of patient discomfort and reduction of insensible water loss, which may decrease the occurrence of dehydration. Risks of lowering fever include delayed identification of the underlying diagnosis and initiation of appropriate treatment and drug toxicity.

There is no evidence that children with fever, as opposed to hyperthermia, are at increased risk of adverse outcomes such as brain damage.^{10,12,24–26} Fever is a common and normal physiologic response that results in an increase in the hypothalamic "set point" in response to endogenous and exogenous pyrogens.^{12,26} In contrast, hyperthermia is a rare and pathophysiologic response with failure of normal homeostasis (no change in the hypothalamic set point) that results in heat production that exceeds the capability to dissipate heat.^{12,26} Characteristics of hyperthermia include hot, dry skin and central nervous system dysfunction that results in delirium, convulsions, or coma.²⁶ Hyperthermia should be addressed promptly, because at temperatures above 41°C to 42°C, adverse physiologic effects begin to occur.^{10,12,27} Studies of health care workers, including physicians, have revealed that most believe that the risk of heat-related adverse outcomes is increased with temperatures above 40°C (104°F), although this belief is not justified.^{7,26,28–30} A child with a temperature of 40°C (104°F) attributable to a simple febrile illness is quite different from a child with a temperature of 40°C (104°F) attributable to heat stroke. Thus, extrapolating similar outcomes from these different illnesses is problematic.

TREATMENT GOALS

A discussion of the use of antipyretics in febrile children must begin with consideration of the therapeutic end points. When counseling families, physicians should emphasize the child's comfort and signs of serious illness rather than emphasizing normothermia. A primary goal of treating the febrile child should be to improve the child's overall

comfort. Most pediatricians observe, with some supporting data from research, that febrile children have altered activity, sleep, and behavior in addition to decreased oral intake.³¹ Unfortunately, there is a paucity of clinical research addressing the extent to which antipyretics improve discomfort associated with fever or illness. It is not clear whether comfort improves with a normalized temperature, because external cooling measures, such as tepid sponge baths, can lower the body temperature without improving comfort.^{10,32} The use of alcohol baths is not an appropriate cooling method, because there have been reported adverse events associated with systemic absorption of alcohol.³³ Furthermore, antipyretics have other clinical outcomes, including analgesia, which may enhance their overall clinical effect. Regardless of the exact mechanism of action, many physicians continue to encourage the use of antipyretics with the belief that most of the benefits are the result of improved comfort and the accompanying improvements in activity and feeding, less irritability, and a more reliable sense of the child's overall clinical condition. Because these are the most important benefits of antipyretic therapy, it is of paramount importance that parental counseling focus on monitoring of activity, observing for signs of serious illness, and appropriate fluid intake to maintain hydration.

The desire to improve the overall comfort of the febrile child must be balanced against the desire to simply lower the body temperature. It is well documented that there are significant concerns on the part of parents, nurses, and physicians about potential adverse effects of fever that have led to a description in the literature of "fever phobia."³⁴ The most consistently identified

serious concern of caregivers and health care providers is that high fevers, if left untreated, are associated with seizures, brain damage, and death.^{1,28,35,36} It is argued that by creating undue concern over these presumed risks of fever, for which there is no clearly established relationship, physicians are promoting an exaggerated desire in parents to achieve normothermia by aggressively treating fever in their children.

There is no evidence that reducing fever reduces morbidity or mortality from a febrile illness. Possible exceptions to this could be children with underlying chronic diseases that may result in limited metabolic reserves or children who are critically ill, because these children may not tolerate the increased metabolic demands of fever.³⁷ Finally, there is no evidence that antipyretic therapy decreases the recurrence of febrile seizures.^{25,38,39}

Despite insufficient evidence, many pediatricians recommend the routine practice of pretreatment with acetaminophen or ibuprofen before a patient receives immunizations to decrease the discomfort associated with the injections and subsequently at the injection sites and to minimize the febrile response.^{12,20,40–42} In addition, results of 1 recent study suggested the possibility of

decreased immune response to vaccines in patients treated early with antipyretics.^{43,44}

Although the available literature is limited on the actual risks of fever and the benefits of antipyretic therapy, it is recognized that improvement in patient comfort is a reasonable therapeutic objective. Furthermore, at this time, there is no evidence that temperature reduction, in and of itself, should be the primary goal of antipyretic therapy.

Acetaminophen

After sufficient evidence emerged of an association between salicylates and Reye syndrome, acetaminophen essentially replaced aspirin as the primary treatment of fever. Acetaminophen doses of 10 to 15 mg/kg per dose given every 4 to 6 hours orally are generally regarded as safe and effective. Typically, the onset of an antipyretic effect is within 30 to 60 minutes; approximately 80% of children will experience a decreased temperature within that time (Table 1).

Although alternative dosing regimens have been suggested,^{45–47} no consistent evidence has indicated that the use of an initial loading dose by either the oral (30 mg/kg per dose) or rectal (40 mg/kg per dose) route improves antipyretic efficacy. The higher rectal dose is often used in intraoperative

TABLE 1 Antipyretic Information

Variable	Acetaminophen	Ibuprofen
Decline in temperature, °C	1–2	1–2
Time to onset, h	<1	<1
Time to peak effect, h	3–4	3–4
Duration of effect, h	4–6	6–8
Dose, mg/kg	10–15 every 4 h	10 every 6 h
Maximum daily dose, mg/kg	90 mg/kg ^a	40 mg/kg
Maximum daily adult dose, g/d	4	2.4
Lower age limit, mo ^b	3	6

Data represent approximate averages from referenced sources.^{46,47,56,58,76,87}

a Label is for 75 mg/kg; 90 mg/kg per day should be limited to less than 3 consecutive days.^{88–90}

b Unless specifically recommended by a health care provider for the younger patient and, then, only after the infant has been examined by a health care provider.

conditions but cannot be recommended for use in routine clinical care.^{48,49} The use of higher loading doses in clinical practice would add potential risks for dosing confusion leading to hepatotoxicity; therefore, such doses are not recommended.

Although hepatotoxicity with acetaminophen at recommended doses has been reported rarely, hepatotoxicity is most commonly seen in the setting of an acute overdose. In addition, there is significant concern over the possibility of acetaminophen-related hepatitis in the setting of a chronic overdose. The most commonly reported scenarios are those of children receiving multiple supratherapeutic doses (ie, >15 mg/kg per dose) or frequent administration of appropriate single doses at intervals of less than 4 hours, which has resulted in doses of more than 90 mg/kg per day for several days. Giving an adult preparation of acetaminophen to a child may result in supratherapeutic dosing. In 1 case series,⁵⁰ half of the children with hepatotoxicity had received adult preparations of acetaminophen.

One safety concern is the effect of acetaminophen on asthma-related symptoms; although asthma has also been associated with acetaminophen use, causality has not been demonstrated.^{52–55}

Ibuprofen

The use of ibuprofen to manage fever has been increasing, because it seems to have a longer clinical effect related to lowering of the body temperature (Table 1). Studies in which the effectiveness of ibuprofen and acetaminophen were compared have yielded variable results; the consensus is that both drugs are more effective than placebo in reducing fever and that ibuprofen (10 mg/kg per dose) is at least as effective as, and perhaps more

effective than, acetaminophen (15 mg/kg per dose) in lowering body temperature when either drug is given as a single or repetitive dose.^{56–62} Data also show that the height of the fever and the age of the child (rather than the specific medication used) may be the primary determinants of the efficacy of antipyretic therapy; those who have a higher fever and are older than 6 years show decreased efficacy or response to antipyretic therapy.⁵⁸ Studies that compare the effect of ibuprofen versus acetaminophen on children's behavior and comfort are generally lacking.

There is no evidence to indicate that there is a significant difference in the safety of standard doses of ibuprofen versus acetaminophen in generally healthy children between 6 months and 12 years of age with febrile illnesses.⁶³ Similar to other nonsteroidal anti-inflammatory drugs (NSAIDs), ibuprofen can potentially cause gastritis,^{64,65} although no data suggest that this is a common occurrence when used on an acute basis, such as during a febrile illness.⁵⁸ However, there have been case reports of bleeding, gastritis, and ulcers of the stomach, duodenum, and esophagus associated with many NSAIDs, including ibuprofen, even when used in typical antipyretic and analgesic doses.^{64,65} Ibuprofen does not seem to worsen asthma symptoms.

Concern has been raised over the nephrotoxicity of ibuprofen. In numerous case reports, children with febrile illnesses developed renal insufficiency when treated with ibuprofen or other NSAIDs. Thus, caution is encouraged when using ibuprofen in children with dehydration or with complex medical illnesses.^{66–68} In children with dehydration, prostaglandin synthesis becomes an increasingly important mechanism for

maintaining appropriate renal blood flow. The use of ibuprofen or any NSAID interferes with the renal effects of prostaglandins, which reduces renal blood flow and potentially precipitates or worsens renal dysfunction.^{66,68} However, it is not possible to determine the actual incidence of ibuprofen-related renal insufficiency after short-term use, because it has not been systematically investigated or reported.⁶⁹ Children who are at greatest risk of ibuprofen-related renal toxicity are those with dehydration, cardiovascular disease, preexisting renal disease, or the concomitant use of other nephrotoxic agents.⁶⁷ Another potential group at risk is infants younger than 6 months because of the possibility of differences in ibuprofen pharmacokinetics and developmental differences in renal function.⁷⁰ Data are inadequate to support a specific recommendation for the use of ibuprofen for fever or pain in infants younger than 6 months (there are dosing data for neonatal closure of patent ductus arteriosus^{71,72}), although the package insert states to “ask a doctor” for guidance on its use in this population. Another potential risk associated with the use of ibuprofen is the possible association between ibuprofen and varicella-related invasive group A streptococcal infection.^{73,74} However, at the time of this report, data were insufficient to support a causal relationship between ibuprofen and invasive group A streptococcal disease.

Alternating or Combination Therapy

A practice frequently used to control fever is the alternating or combined use of acetaminophen and ibuprofen. In a convenience sample survey of 256 parents or caregivers, 67% reported alternating acetaminophen and ibuprofen for fever control, 81% of whom stated

that they had followed the advice of their health care provider or pediatrician.⁷⁵ Although 4 hours was the most frequent interval, parents reported alternating therapy every 2, 3, 4, and 6 hours, which suggests that there is no consensus on dosing instructions.

At the time of this report, 5 studies had been identified that compared alternating ibuprofen and acetaminophen versus either acetaminophen or ibuprofen as single agents.^{76–80} Initially, changes in temperature were similar for all groups in these studies, regardless of therapy. However, 4 or more hours after the initiation of treatment, lower temperature was consistently observed in the combination-treatment groups. For example, 6 and 8 hours after the initiation of the study, a greater percentage of children were afebrile in the combination group (83% and 81%, respectively) compared with those in the group that received ibuprofen alone (58% and 35%, respectively).⁷¹ Only 1 study⁷⁷ evaluated issues related to stress and comfort and found lower stress scores and less time missed from child care in the combination-treatment group. Another study⁷⁸ showed a trend toward a normalization of fever-related symptoms by 24 and 48 hours after institution of therapy, but these trends disappeared by day 5.

Although the aforementioned studies provide some evidence that combination therapy may be more effective at lowering temperature, questions remain regarding the safety of this practice as well as the effectiveness in improving discomfort, which is the primary treatment end point. The possibility that parents will either not receive or not understand dosing instructions, combined with the wide array of formulations that contain these drugs, increases the

potential for inaccurate dosing or overdosing.^{81,82} Finally, this practice may only promote the fever phobia that already exists.

Although there is some evidence that combination therapy may result in a lower body temperature for a greater period of time, there is no evidence that combination therapy results in overall improvement in other clinical outcomes. Also, these studies have not contained adequate numbers of subjects to fully evaluate the safety of this practice. Therefore, there is insufficient evidence to support or refute the routine use of combination treatment with both acetaminophen and ibuprofen. Practitioners who choose to follow this practice should counsel parents carefully regarding proper formulation, dosing, and dosing intervals and emphasize the child's comfort instead of reduction of fever.

INSTRUCTIONS FOR CAREGIVERS

It is critically important for pediatricians to clearly describe the appropriate use (ie, formulation, dose, and dosing interval) of acetaminophen and ibuprofen to caregivers (Table 1). Child safety will be further enhanced by clear labeling and the development of simplified dosing methods, standardized drug concentrations, and standardized delivery devices.^{83–85} Cough-and-cold products that contain acetaminophen and ibuprofen should not be given to children because of the possibility that parents may unintentionally give their child simultaneous doses of an antipyretic and a cough-and-cold medication that contains the same antipyretic. In addition, there is a lack of proven efficacy for this class of combination products for children. For children who require liquid preparations, physicians should encourage families to only

use 1 formulation. Acetaminophen is the most common single ingredient implicated in emergency department visits for medication overdoses among children, and more than 80% of these emergency visits are a result of unsupervised ingestions⁸⁶; therefore, proper handling and storage of antipyretics should be encouraged.

SUMMARY

Appropriate counseling on the management of fever begins by helping parents understand that fever, in and of itself, is not known to endanger a generally healthy child. In contrast, fever may actually be of benefit; thus, the real goal of antipyretic therapy is not simply to normalize body temperature but to improve the overall comfort and well-being of the child. Acetaminophen and ibuprofen, when used in appropriate doses, are generally regarded as safe and effective agents in most clinical situations. However, as with all drugs, they should be used judiciously to minimize the risk of adverse drug effects and toxicity. Combination therapy with acetaminophen and ibuprofen may place infants and children at increased risk because of dosing errors and adverse outcomes, and these potential risks must be carefully considered. When counseling a family on the management of fever in a child, pediatricians and other health care providers should minimize fever phobia and emphasize that antipyretic use does not prevent febrile seizures. Pediatricians should focus instead on monitoring for signs/symptoms of serious illness, improving the child's comfort by maintaining hydration, and educating parents on the appropriate use, dosing, and safe storage of antipyretics. To promote child safety, pediatricians should advocate for a limited number of

formulations of acetaminophen and ibuprofen and for clear labeling of dosing instructions and an included dosing device for antipyretic products.

LEAD AUTHORS

Janice E. Sullivan, MD
Henry C. Farrar, MD

COMMITTEE ON DRUGS, 2009–2010

Daniel A. C. Frattarelli, MD,
Chairperson
Jeffrey L. Galinkin, MD
Thomas P. Green, MD
Mary A. Hegenbarth, MD
Mark L. Hudak, MD
Matthew E. Knight, MD
Robert E. Shaddy, MD

FORMER COMMITTEE ON DRUGS MEMBER

Wayne R. Snodgrass, MD, PhD

CONSULTANT

Robert M. Ward, MD

LIAISONS

John J. Alexander, MD – *Food and Drug Administration*
Janet D. Cragan, MD – *Centers for Disease Control and Prevention*
George P. Giacoia, MD – *National Institutes of Health*
Michael J. Rieder, MD – *Canadian Paediatric Society*
Adelaide Robb, MD – *American Academy of Child and Adolescent Psychiatry*
Hari C. Sachs, MD – *Food and Drug Administration*

STAFF

Raymond J. Koteras, MHA
rkoteras@aap.org

SECTION ON CLINICAL PHARMACOLOGY AND THERAPEUTICS, 2009–2010

Janice E. Sullivan, MD, Chairperson
Glen S. Frick, MD
Lynne G. Maxwell, MD
Ian M. Paul, MD

John F. Pope, MD
Thomas G. Wells, MD

FORMER EXECUTIVE COMMITTEE MEMBERS

Charles J. Cote, MD
Henry C. Farrar, MD
Richard L. Gorman, MD

STAFF

Raymond J. Koteras, MHA

2022 REFERENCE UPDATE ACKNOWLEDGMENT

John W. Berkenbosch, MD, FAAP,
FCCM

ABBREVIATION

NSAID: nonsteroidal anti-inflammatory drug

REFERENCES

1. Crocetti M, Moghbeli N, Serwint J. Fever phobia revisited: have parental misconceptions about fever changed in 20 years. *Pediatrics*. 2001;107(6):1241–1246
2. Bilenko N, Tessler H, Okbe R, Press J, Gorodischer R. Determinants of antipyretic misuse in children up to 5 years of age: a cross-sectional study. *Clin Ther*. 2006;28(5):783–793
3. Kramer MS, Naimark L, Leduc DG. Parental fever phobia and its correlates. *Pediatrics*. 1985;75(6):1110–1113
4. Hiller MG, Caffery MS, Bégué RE. A Survey About Fever Knowledge, Attitudes, and Practices Among Parents. *Clin Pediatr (Phila)*. 2019;58:677–680
5. Li SF, Lacher B, Crain EF. Acetaminophen and ibuprofen dosing by parents. *Pediatr Emerg Care*. 2000;16(6):394–397
6. Chiappini E, Cangelosi AM, Becherucci P, et al. Knowledge, attitudes, and misconceptions of Italian healthcare professionals regarding fever management in children. *BMC Pediatr*. 2018;18:194

7. May A, Bauchner H. Fever phobia: the pediatrician's contribution. *Pediatrics*. 1992;90(6):851–854
8. Mayoral CE, Marino RV, Rosenfeld W, Greensher J. Alternating antipyretics: is this an alternative? *Pediatrics*. 2000;105(5):1009–1012
9. El-Radhi AS. Why is the evidence not affecting the practice of fever management? *Arch Dis Child*. 2008;93(11):918–920
10. Peetoom KK, Smits JJ, Ploum LJ, et al. Does well-child care education improve consultations and medication management for childhood fever and common infections? A systematic review. *Arch Dis Child*. 2017;102:261–267
11. Jaffe DM. Assessment of the child with fever. In: Rudolph CD, Rudolph AM, Hostetter MK, Lister GE, Siegel NJ eds. *Rudolph's Pediatrics*. 21st ed. New York, NY: McGraw-Hill; 2002:302–309
12. Kohl KS, Marcy SM, Blum M, et al; Brighton Collaboration Fever Working Group. Fever after immunization: current concepts and improved future scientific understanding. *Clin Infect Dis*. 2004;39(3):389–394
13. Hasday JD, Garrison A. Antipyretic therapy in patients with sepsis. *Clin Infect Dis*. 2000;31(suppl 5):S234–S241
14. Adam HM. Fever and host responses. *Pediatr Rev*. 1996;17(9):330–331
15. Kluger MJ. Fever revisited. *Pediatrics*. 1992;90(6):846–850
16. Kluger MJ. Fever: role of pyrogens and cryogens. *Physiol Rev*. 1991;71(1):93–127
17. Roberts NJ. Impact of temperature elevation on immunologic defenses. *Rev Infect Dis*. 1991;13(3):462–272
18. Nizet V, Vinci RJ, Lovejoy FH. Fever in children. *Pediatr Rev*. 1994;15(4):127–135
19. Doran TF, De Angelis C, Baumgardner RA, Mellits ED. Acetaminophen: more harm than good for chickenpox? *J Pediatr*. 1989;114(6):1045–1048
20. Michael Marcy S, Kohl KS, Dagan R, et al; Brighton Collaboration Fever Working Group. Fever as an adverse event following immunization: case definition and guidelines of data collection,

- analysis and presentation. *Vaccine*. 2004;22(5–6):551–556
21. Plaisance KI, Kudravalli S, Wasserman SS, Levine MM, Mackowiak PA. Effect of antipyretic therapy on the duration of illness in experimental influenza A, *Shigella sonnei*, and *Rickettsia rickettsii* infections. *Pharmacotherapy*. 2000; 20(12):1417–1422
 22. Burnett AM, Domachowske JB. Therapeutic considerations for children with invasive group A streptococcal infections: a case series report and review of the literature. *Clin Pediatr (Phila)*. 2007;46(6):550–555
 23. Ospina CAC, Salcedo A. Ibuprofen increases soft tissue infections in children. *BMJ*. 2008;337:a1767
 24. Schmitt BD. Fever in childhood. *Pediatrics*. 1984;74(5 pt 2):929–936
 25. American Academy of Pediatrics, Steering Committee on Quality Improvement and Management, Subcommittee on Febrile Seizures. Febrile seizures: clinical practice guidelines for the long-term management of the child with simple febrile seizures. *Pediatrics*. 2008;121(6):1281–1286
 26. Bouchama A, Knochel JP. Heat Stroke. *N Engl J Med*. 2002;346(25):1978–1988
 27. Trautner BW, Caviness AC, Gerlach GR, Demmler G, Macias CG. Prospective evaluation of the risk of serious bacterial infection in children who present to the emergency department with hyperpyrexia (temperature of 106°F or higher). *Pediatrics*. 2006;118(1):34–40
 28. Poirier MP, Davis PH, Gonzalez Del Ray JA, Monroe KW. Pediatric emergency department nurses' perspective on fever in children. *Pediatr Emerg Care*. 2000;16(1):9–12
 29. Howe AS, Boden BP. Heat-related illness in athletes. *Am J Sports Med*. 2007; 35(8):1384–1395
 30. Jardine DS. Heat illness and heat stroke. *Pediatr Rev*. 2007;28(7):249–258
 31. Mistry RD, Stevens MW, Gorelick MH. Short-term outcomes of pediatric emergency department febrile illnesses. *Pediatr Emerg Care*. 2007;23(9): 617–623
 32. Greisman LA, Mackowiak PA. Fever: beneficial and detrimental effects of antipyretics. *Curr Opin Infect Dis*. 2002;15(3):241–245
 33. Meremikwu M, Oyo-Ita A. Physical methods versus drug placebo or no treatment for managing fever in children. *Cochrane Database Syst Rev*. 2003; (2):CD004264
 34. Schmitt BD. Fever phobia. *Am J Dis Child*. 1980;134(2):176–181
 35. Betz MG, Grunfeld AF. Fever phobia in the emergency department: a survey of children's caregivers. *Eur J Emerg Med*. 2006;13(3):129–133
 36. Karwowska A, Nijssen-Jordan C, Johnson D, Davies HD. Parental and health care provider understanding of childhood fever: a Canadian perspective. *CJEM*. 2002;4(6):394–400
 37. Kayman H. Management of fever: making evidence-based decisions. *Clin Pediatr (Phila)*. 2003;42(5):383–392
 38. Duffner PK, Baumann RJ. A synopsis of the American Academy of Pediatrics practice parameters on the evaluation and treatment of children with febrile seizures. *Pediatr Rev*. 1999;20(8): 285–287
 39. Sadleir LG, Scheffer IE. Febrile seizures. *BMJ*. 2007;334(7588):307–311
 40. Lewis K, Cherry JD, Sachs MH, Tarle JM, Overturf GD. The effect of prophylactic acetaminophen administration on reactions to DTP vaccination. *Am J Dis Child*. 1988;142(1):62–65
 41. Ipp MM, Gold R, Greenberg S, et al. Acetaminophen prophylaxis of adverse reactions following vaccination of infants with diphtheria-pertussis-tetanus toxoids-polio vaccine. *Pediatr Infect Dis J*. 1987;6(8):721–725
 42. Centers for Disease Control and Prevention. Pertussis vaccination: use of acellular pertussis vaccines among infants and children—recommendations of the Advisory Committee on Immunization Practices (ACIP) [published correction appears in *MMWR Morb Mortal Wkly Rep*. 1997;46(30):706]. *MMWR Recomm Rep*. 1997;46(RR-7):1–25. Available at: www.cdc.gov/mmwr/preview/mmwrhtml/00048610.htm. Accessed October 13, 2009
 43. Prymular, Siegrist CA, Chlibeck R, et al. Effect of prophylactic paracetamol administration at time of vaccination on febrile reactions and antibody responses in children: two open-label, randomized controlled trials. *Lancet*. 2009;374(9698):1339–1350
 44. Wysocki J, Center KJ, Brzostek J, et al. A randomized study of fever prophylaxis and the immunogenicity of routine pediatric vaccinations. *Vaccine*. 2017;35:1926–1935
 45. Tréluyer JM, Tonnelier S, d'Anthis P, Leclerc B, Jolivet-Landreau I, Pons G. Antipyretic efficacy of an initial 30 mg/kg loading dose of acetaminophen versus a 15 mg/kg maintenance dose. *Pediatrics*. 2001;108(4). Available at: www.pediatrics.org/cgi/content/full/108/4/e73
 46. Nabulsi M, Tamim H, Sabra R, et al. Equal antipyretic effectiveness of oral and rectal acetaminophen: a randomized controlled trial. *BMC Pediatr*. 2005;5:35–42
 47. Scolnik D, Kozer E, Jacobson S, Diamond S, Young NL. Comparison of oral versus normal and high dose rectal acetaminophen in the treatment of febrile children. *Pediatrics*. 2002; 110(3):553–556
 48. Birmingham PK, Tobin MJ, Fisher DM, Henthorn TK, Hall SC, Coté CJ. Initial and subsequent dosing of rectal acetaminophen in children. *Anesthesiology*. 2001;94(3):385–389
 49. Birmingham PK, Tobin MJ, Henthorn TK, et al. Twenty-four-hour pharmacokinetics of rectal acetaminophen in children: an old drug with new recommendations. *Anesthesiology*. 1997;87(2): 244–252
 50. Heubi JE, Barbacci M, Zimmerman HJ. Therapeutic misadventures with acetaminophen: hepatotoxicity after multiple doses in children. *J Pediatr*. 1998; 132(1):22–27
 51. Henretiz FM, Selbst SM, Forrest C, Kearney TK, Orel H, Werner S, Williams TA. Repeated acetaminophen overdosing: causing hepatotoxicity in children—clinical reports and literature review. *Clin Pediatr (Phila)*. 1989;28(11): 525–528
 52. Kanabar D, Dale S, Rawat M. A review of ibuprofen and acetaminophen use in febrile children and the occurrence of asthma-related symptoms. *Clin Ther*. 2007;29(12):2716–2723

53. Lesko SM, Louik C, Vezina RM, Mitchell AA. Asthma morbidity after short-term use of ibuprofen in children. *Pediatrics*. 2002;109(2). Available at: www.pediatrics.org/cgi/content/full/109/2/e20
54. Etminan M, Sadasafavi M, Jafari S, Doyle-Waters M, Aminzadeh K, Fitzgerald JM. Acetaminophen use and the risk of asthma in children and adults: a systematic review and meta-analysis. *Chest*. 2009;136(5):1316–1323
55. Allmers H, Skudlik C, John SM. Acetaminophen use: a risk for asthma? *Curr Allergy Asthma Rep*. 2009;9(2):164–167
56. Goldman RD, Ko K, Linett LJ, Scolnik D. Antipyretic efficacy and safety of ibuprofen and acetaminophen in children. *Ann Pharmacother*. 2004;38(1):146–150
57. Perrott DA, Piira T, Goodenough B, Champion D. Efficacy and safety of acetaminophen vs ibuprofen for treating children's pain or fever: a meta-analysis. *Arch Pediatr Adolesc Med*. 2004;158(6):521–526
58. Wilson JT, Brown RD, Kearns GL, et al. Single-dose, placebo-controlled comparative study of ibuprofen and acetaminophen antipyresis in children. *J Pediatr*. 1991;119(5):803–811
59. Walson PD, Galletta G, Chomilo F, Braden NJ, Sawyer LA, Scheinbaum ML. Comparison of multidose ibuprofen and acetaminophen therapy in febrile children. *Am J Dis Child*. 1992;146(5):626–632
60. Kauffman RE, Sawyer LA, Scheinbaum ML. Antipyretic efficacy of ibuprofen vs acetaminophen. *Am J Dis Child*. 1992;146(5):622–625
61. Van Esch A, Van Steensel-Moll HA, Steyerberg EW, Offringa M, Habbema JD, Derksen-Lubsen G. Antipyretic efficacy of ibuprofen and acetaminophen in children with febrile seizures. *Arch Pediatr Adolesc Med*. 1995;149(6):632–637
62. **Narayan K, Cooper S, Morphet J, Innes K. Effectiveness of paracetamol versus ibuprofen administration in febrile children: A systematic literature review. *J Paediatr Child Health*. 2017;53:800–807**
63. Lesko SM, Mitchell AA. The safety of acetaminophen and ibuprofen among children younger than two years of age. *Pediatrics*. 1999;104(4). Available at: www.pediatrics.org/cgi/content/full/104/4/e39
64. Autret-Leca E, Bensouda-Grimaldi L, Maurage C, Jonville-Bera AP. Upper gastrointestinal complications associated with NSAID's in children[in French]. *Therapie*. 2007;62(2):173–176
65. Berezin SH, Bostwick HE, Halata MS, Feerick J, Newman LJ, Medow MS. Gastrointestinal bleeding in children following ingestion of low dose ibuprofen. *J Pediatr Gastroenterol Nutr*. 2007;44(4):506–508
66. Ulinski T, Guigonis V, Dunan O. Acute renal failure after treatment with non-steroidal anti-inflammatory drugs. *Eur J Pediatr*. 2004;163(3):148–150
67. John CM, Shukla R, Jones CA. Using NSAID in volume depleted children can precipitate acute renal failure. *Arch Dis Child*. 2007;92(6):524–526
68. Moghal NE, Hegde S, Eastham KM. Ibuprofen and acute renal failure in a toddler. *Arch Dis Child*. 2004;89(3):276–277
69. Lesko SM, Mitchell AA. Renal function after short-term ibuprofen use in infants and children. *Pediatrics*. 1997;100(6):954–957
70. Allegaert K, Cossey V, Debeer A, et al. The impact of ibuprofen on renal clearance in preterm infants is independent of the gestational age. *Pediatr Nephrol*. 2005;20(6):740–743
71. Hammerman C, Shchors I, Jacobson S, Schimmel MS, Bromiker R, Kaplan M, Nir A. Ibuprofen versus continuous indomethacin in premature neonates with patent ductus arteriosus: is the difference in the mode of administration? *Pediatr Res*. 2008;64(3):291
72. Su BH, Lin HC, Chiu HY, Hsieh HY, Chen HH, Tsai YC. Comparison of ibuprofen and indomethacin for early-targeted treatment of patent ductus arteriosus in extremely premature infants: a randomized controlled trial. *Arch Dis Child Fetal Neonatal Ed*. 2008;93(2):F94–F99
73. Zerr DM, Alexander ER, Duchin JS, Koutsky LA, Rubens CE. A case-control study of necrotizing fasciitis during primary varicella. *Pediatrics*. 1999;103(4 pt 1):783–790
74. Lesko SM, O'Brien KL, Schwartz B, Vezina R, Mitchell AA. Invasive group A streptococcal infection and nonsteroidal anti-inflammatory drug use among children with primary varicella. *Pediatrics*. 2001;107(5):1108–1115
75. Wright AD, Liebelt EL. Alternating antipyretics for fever reduction in children: an unfounded practice passed down to parents from pediatricians. *Clin Pediatr (Phila)*. 2007;46(2):146–150
76. Nabulsi MM, Tamim H, Mahfoud Z, et al. Alternating ibuprofen and acetaminophen in the treatment of febrile children: a pilot study. *BMC Med*. 2006;4:4–12
77. Sarrell EM, Wielunsky E, Cohen HA. Antipyretic treatment in young children with fever: acetaminophen, ibuprofen or both alternating in a randomized double-blind study. *Arch Pediatr Adolesc Med*. 2006;160(2):197–202
78. Hay AD, Costelloe C, Redmond NM, et al. Paracetamol Plus Ibuprofen for the Treatment of Fever in Children (PITCH): randomized controlled trial [published correction appears in *BMJ*. 2009;339:b3295]. *BMJ*. 2008;337:a1302
79. Erlewyn-Lajeunesse MDS, Coppens K, Hunt LP, et al. Randomised controlled trial of combined paracetamol and ibuprofen for fever. *Arch Dis Child*. 2006;91(5):414–416
80. Kramer LC, Richards PA, Thompson AM, Harper DP, Fairchok MP. Alternating antipyretics: antipyretic efficacy of acetaminophen versus acetaminophen alternated with ibuprofen in children. *Clin Pediatr (Phila)*. 2008;47(9):907–911
81. Schmitt BD. Concerns over alternating acetaminophen and ibuprofen for fever. *Arch Pediatr Adolesc Med*. 2006;160(7):757
82. Saphyakhajon P, Greene G. Alternating acetaminophen and ibuprofen in children may cause parental confusion and is dangerous. *Arch Pediatr Adolesc Med*. 2006;160(7):757
83. Frush KS, Lao X, Hutchinson P, Higgins JN. Evaluation of a method to reduce over-the-counter medication dosing error. *Arch Pediatr Adolesc Med*. 2004;158(7):620–624
84. Rand CM, Conn KM, Crittenden CN, Halterman JS. Does a color-coded

- method for measuring acetaminophen doses reduce the likelihood of dosing error? *Arch Pediatr Adolesc Med*. 2004;158(7):625–627
85. Food and Drug Administration, Center for Drug Evaluation and Research. Joint meeting of the Drug Safety and Risk Management Advisory Committee, Nonprescription Drugs Advisory Committee, and the Anesthetic and Life Support Drugs Advisory Committee; June 29–30, 2009: questions to the committee. Available at: www.fda.gov/downloads/AdvisoryCommittees/CommitteesMeetingMaterials/Drugs/DrugSafetyandRiskManagementAdvisoryCommittee/UCM170188.pdf. Accessed May 12, 2010
 86. Schillie SF, Shehab N, Thomas KE, Budnitz DS. Medication overdoses leading to emergency department visits among children. *Am J Prev Med*. 2009;37(3):181–187
 87. Robertson J, Shilkofski N. *The Harriet Lane Handbook: Formulary*. 17th ed. Philadelphia, PA: Elsevier/Mosby; 2005
 88. Temple AR. Pediatric dosing of acetaminophen. *Pediatr Pharmacol (New York)*. 1983;3(3–4):321–327
 89. Dart RC, Erdman AR, Olson KR, et al; American Association of Poison Control Centers. Acetaminophen poisoning: an evidence-based consensus guideline for out-of-hospital management. *Clin Toxicol (Phila)*. 2006;44(1):1–18
 90. Penna AC, Sawson KP, Penna CM. Is prescribing paracetamol “pro re nata” acceptable? *J Paediatr Child Health*. 1993;29(2):104–106